Eleryc

Carbon dioxide removal prepurchase application
2025

General Application

(The General Application applies to everyone; all applicants should complete this)

Public section

The content in this section (answers to questions 1(a) - (d)) will be made public on the Frontier GitHub repository after we announce purchases in the summer and fall. Include as much detail as possible but omit sensitive and proprietary information.

Company or organization name

Eleryc Corporation

Company or organization location (we welcome applicants from anywhere in the world)

1911 Lundy Avenue
San Jose, CA 95131

Name(s) of primary point(s) of contact for this application

Ian Robinson, CEO

Brief company or organization description <20 words

Funded by leading climate investors, Eleryc is dedicated to lowering carbon concentrations in the atmosphere through a unique, low-energy electrolysis process.

1. Public summary of proposed project to Frontier

a. **Description of the CDR approach:** Describe how the proposed technology removes CO_2 from the atmosphere, including how the carbon is stored for > 1,000 years. Tell us why your system is best-in-class, and how you're differentiated from any other organization working on a similar approach. If your project addresses any of the priority innovation areas identified in the RFP, tell us how. Please include figures and system schematics and be specific, but concise. 1000-1500 words

Eleryc's process technology leverages simple acid-base chemistry to facilitate the removal of CO_2 from the atmosphere via direct air capture (DAC). The process is:

- Simple: all process streams are gas streams or aqueous liquids that can easily be transported by pumps and blowers.
- Circular: the overall process takes in air and produces a stream of CO_2 depleted air and purified CO_2. There is a small amount of make-up water to balance water losses into the CO_2 depleted air. Otherwise, the system is self-contained.

- Robust: The process equipment is standard across a wide range of industries and is readily available and easily scaled via modularization.
- Flexible: The CO_2 capture rate can be adjusted in real time to respond to the availability of renewables on the grid.
- Low Cost: Third-party studies have validated Eleryc's cost models showing that the N^{th} of a kind (NOAK) plants can easily achieve levelized costs below $100/tonne CO_2.

As shown below in Figure 1, an electrolyzer produces a strong base (caustic) used to capture and convert atmospheric CO_2 into carbonate ions, which are subsequently reacted with acid (also produced in the electrolyzer) to release purified CO_2 for permanent storage in underground geologic reservoirs.

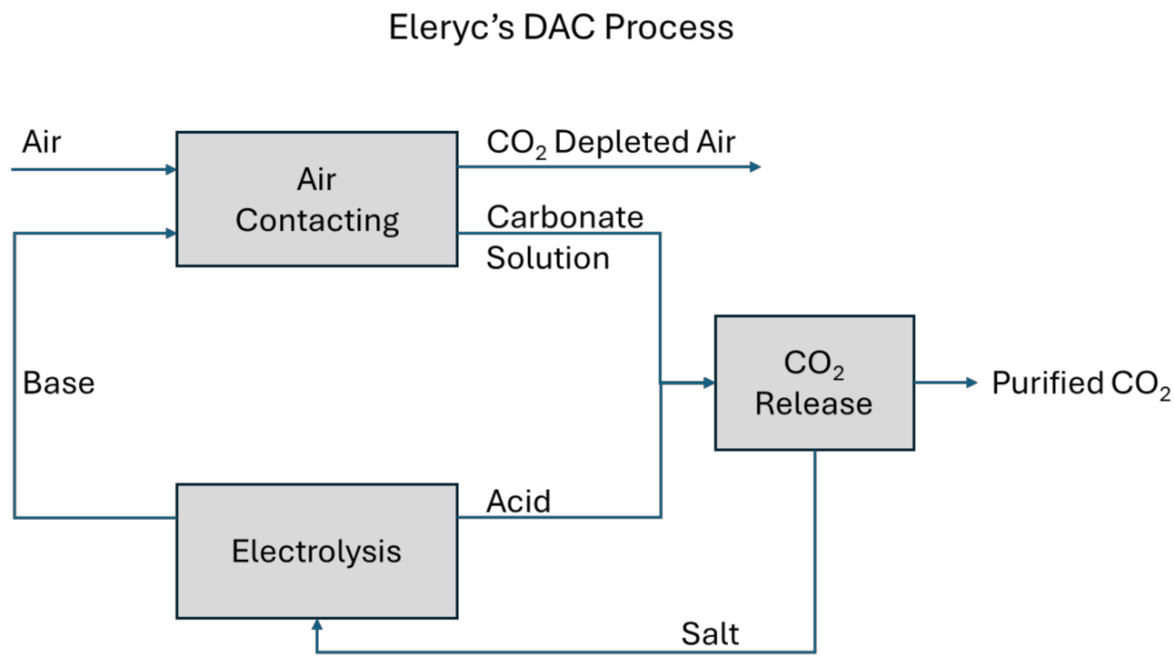

Figure 1: Block Flow Diagram of Eleryc's DAC Process

Eleryc's process achieves best-in-class cost of capture and energy demand through three key features. First, Eleryc's proprietary chemistry enables low-energy generation of acid and base by combining a novel redox couple (with low voltage standard potentials) at the anode and cathode with (non-chloride containing) salts selected to minimize the overall cost of capture. Second, the capture solution is a stream of aqueous caustic, which is selective for CO_2 even in humid environments, has the highest rate of mass transfer, minimizing contactor footprint, and alleviates the environmental and energy concerns associated with amine capture and thermal regeneration. Third, Eleryc's process liberates CO_2 chemically via acidification which reduces the total process energy requirements, produces a very pure CO_2 stream, and leverages the use of renewable power.

The heart of the system is an electrolysis unit which generates caustic (base) at cell voltages one-third less than standard chlor-alkali cells. To achieve these voltages, a novel chemistry change at the anode replaces chlorine generation with a proprietary half-cell reaction that reduces the open-circuit potential requirements and eliminates the chloride ions from the process, resulting in a more benign capture process, reduced local community impacts, and lower capital cost.[1]

The base produced at the cathode is fed to the air-contactor where CO_2 is removed from the

[1] The cathode reaction is standard hydrogen evolution reaction (HER).

atmosphere and converted into soluble carbonate ions ($CO_3^{=}$). As noted previously, the use of caustic enables high mass-transfer rates in the air contactor, minimizing the required footprint of the overall plant. In addition, the use of caustic eliminates concerns of amine-based sorbents, which require thermal regeneration from a heat source and are prone to degradation into volatile species that leave with the exhaust air. Use of caustic also avoids the pitfalls of solid-based sorbent systems which also require thermal regeneration and solids handling equipment. These challenges with solid sorbent systems are compounded by the fact that solid sorbents are not selective to CO_2 and will readily absorb water which increases thermal regeneration requirements significantly beyond the requirements to liberate CO_2.

The aqueous carbonate stream recovered from Eleryc's air contactor is pumped to the CO_2 release unit, where carbonate ions react with the acid produced at the anode of the electrolyzer converting the carbonate ions to vapor phase CO_2 which is released as a relatively pure vapor stream. The conversion of the carbonate ions also regenerates the salt fed to the electrolyzer so that the entire process is self-contained – only a small amount of make-up water is required to replace the humidified air leaving the air contactor.

Thus, the overall process is continuous and closed loop with respect to all species except for air and CO_2. No other chemicals cross the process boundary. To water balance the overall process, the small amount of water that enters the CO_2 release unit with the carbonate ions, must be transferred back to the catholyte loop, where it serves as dilution water. This water transfer is accomplished via reverse osmosis (RO).[2]

The CO_2 liberated from the carbonate stream is then compressed and transported for permanent disposition via geologic storage in a Class VI well. Geologic storage in Class VI wells remains the preferred option for DAC technologies at scale because they currently offer the best combination of permanence, capacity and cost and the rigor of reservoir monitoring ensures that the CO_2 lost from the formation will be *de minimis*. (Obviously, if better options emerge, Eleryc would be happy to explore those).

The proposed project is initial capture of 1000 tonnes of CO_2 from a 100,000 tonne per annum (TPA) DAC facility located in the US, but the specific site is currently undetermined. Our site selection process will involve several factors including impact on the local community and other environmental justice considerations, availability of clean or renewable power in the local grid, proximity to a Class VI injection site or access to a Class VI injection site via pipeline and possible incentives. Post site-selection, a rigorous life-cycle analysis (LCA), including Scope 1, Scope 2, and Scope 3 emissions, will ensure that the process results in a net reduction of CO_2 in the atmosphere as calculated. That LCA will subsequently be validated by a state-of-the-art Measurement, Reporting and Verification (MRV) protocol.

Eleryc has completed lab scale testing and has recently begun testing of a prototype cell at our first facility in Campbell, CA. Additionally, we recently acquired an 45,000 sq. ft. facility in San Jose that will serve as the location of our pilot plant, which will de-risk all key technical elements of the technology allowing us to move on to the scale of the proposed project.

Organizationally, Eleryc possesses a depth of relevant development experience and a broad patent portfolio (both in-licensed and internally developed). The company is staffed with experienced professionals from high technology companies such as Tesla and Apple, who have specialized in mass-produced electrochemical systems, clean-tech companies, such as Chemetry and C-Zero, and investment entities (Khosla Ventures and ARPA-E).

[2] Water-balancing is actually slightly more complicated than what is stated here because water will equilibrate in the air contactor between the caustic solution and the exiting air stream. The details of this portion of the balance, which depends on the relative humidity of the incoming stream, are beyond this initial description, but have been carefully reviewed by Eleryc's process engineering team and do not present significant risks to deployment.

b. **Project objectives:** What are you trying to build? Discuss location(s) and scale. What is the current cost breakdown, and what needs to happen for your CDR solution to approach Frontier's cost and scale criteria? What is your approach to quantifying the carbon removed? Please include figures and system schematics and be specific, but concise. 1000-1500 words

As shown below in Figure 2, our plan is to scale from pre-pilot to commercial over the next four years. Our confidence in this rapid timeline is based on the quality of the team, the simplicity of system scaling, the leveraging of existing commercial components, and the experience we have in bringing similar electrochemical technologies to commercial scales.

We have recently commissioned our Single Cell Pre-Pilot unit at our first facility in Campbell, CA. This system was built to test full height electrolyzer cells under realistic hydraulics.[3] We expect testing of cell designs and components to continue through the end of 2026 with the latter portion of this testing primarily aiding in troubleshooting pilot plant cells.

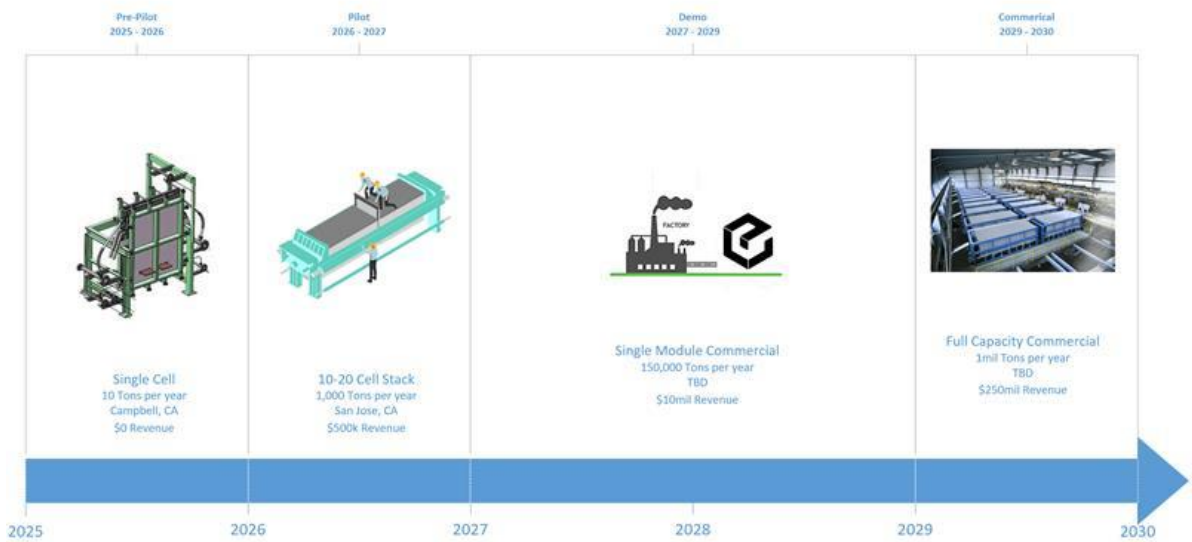

Figure 2: Eleryc's estimated timeline from laboratory results (mostly in 2024) to full scale Commercial.

As the next step in scaling our technology, we are building a 1000 tonne per annum (TPA) Pilot Plant (the Pilot) at a recently acquired facility in San Jose, CA. The Pilot, which is scheduled to be operational in the latter part 2026, will incorporate a stack of 10-20 cells integrated with an air contactor, CO_2 release reactor, and water balancing equipment all controlled through a programmable logic controller (PLC). The Pilot electrolyzer design will incorporate all the features of a full commercial electrolyzer including full width and full height cells, but with fewer cells per stack. The stack will also be of sufficient size to resolve potential full-stack issues such as heat dissipation or contact resistance losses. Although the CO_2 Release reactor will produce a substantially pure CO_2 stream that will be ready for transportation and geological storage, Eleryc's current plan is to demonstrate all the elements up to compression, with compression being an optional addition if

[3] The use of full-height cells is important as the pressure differences across cell components are a function of dynamic head and static liquid head. Thus, the full-height cells, which have identical flows and static heads, permit testing cell hydraulics under commercial conditions. In this way, testing full-height cells simplifies the scale up process which increases simply a matter of increasing the cell width by a factor of 5-7 and numbering up the cells into electrolyzer stacks.

opportunities develop to sequester or use the CO_2 from this facility.[4] Pilot operation will provide confirmation of all critical performance data including recycle streams and advance design readiness to Demonstration scale (the Demo). The successful operation of the Pilot will validate all key process parameters including energy intensity, Faradaic efficiency (FE), component lifetimes, mass transfer coefficients, reaction rates, and CO_2 purity. The resulting data from the Pilot will also be used to update the TEA, LCA and MRV protocols before scaling to the Demonstration plant. These activities will confirm our confidence in our ability to meet Frontier's cost and scale criteria.[5]

Upon successful operation of the Pilot, the process technology and equipment will be sufficiently de-risked to allow design and construction of a first-of-a-kind (FOAK) commercial scale plant (the Demo), which is expected to be nominally 100,000 TPA of gross sequestered CO_2 from the atmosphere. Initial capture and storage from this Demo plant is the proposed project for delivery of pre-purchased CO_2 from Frontier. The location of this plant, which will be the source of the CDR, has not yet been determined. The Demo plant will include Direct Air Capture, CO_2 transportation, and geologic storage. At this point, Eleryc expects to have commercial arrangements with geologic storage providers such as Carbon Terravault (https://www.crc.com/carbon-terravault) or similar that will result in net CDR. Eleryc anticipates the siting of this facility will be based on the following criteria:

- Access to geologic storage to ensure permanent, verifiable CO_2 storage
- Availability of clean or renewable power to ensure the maximum net CO_2 removal
- Provisions for environmental justice and engagement with the adjacent community including maximizing value creation for under-served and under-resourced populations.

As shown below in Figure 3, options for geological storage in the US are growing rapidly. Eleryc expects to situate this Demo plant in such a way that there is access to permanent geological storage of CO_2. The Measurement, Reporting, and Verification (MRV) protocol and Life Cycle Analysis (LCA) will be modified appropriately if the CO_2 is transmitted via pipeline to a common storage location vs. a dedicated site (see below). Also, Eleryc intends to engage a community engagement program with local neighborhoods that may be impacted by the construction and operation of a DAC facility to ensure that local populations are not adversely affected.

Initial cost estimates for an N^{th}-of-a-kind (NOAK) scale plant, based on a full Techno-Economic Analysis (TEA) developed by Eleryc and reviewed by a 3rd party indicate that at full scale, nominally 1M TPA, levelized cost of capture was below the $100/tonne target set by Frontier.[6] Our confidence in this number is high provided we can get access to relatively low-cost clean or renewable power. To enable further scaling, Eleryc will construct a cell manufacturing facility of a size that is on par with existing large-scale battery facilities. This facility would enable Eleryc to deploy facilities at 100-300 DAC injection sites with each facility containing 2-3 plants of 1M TPA scale. That deployment would allow Eleryc's technology to achieve 0.5 GTA of CDR.

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[4] As a unit operation, multi-stage compression in CO_2 service with both centrifugal and reciprocating machines is extremely well understood, rendering this step very low risk. The more important portion of the project will be confirming CO_2 concentrations and water content in the stream feeding the compression.

[5] Our current TEA analysis based on existing lab results and confirmed by a 3rd party show that we will be able to achieve Frontier's cost targets without additional improvements required. Successful operation of the Pilot will also confirm our ability to scale the process as the in-house manufacturing processes and supply chains will have to be confirmed at this point.

[6] 3rd party valuation performed by Strategic Analysis, Inc. for a NOAK plant at 1M TPA capacity. Additional details provided in Section 6 below.

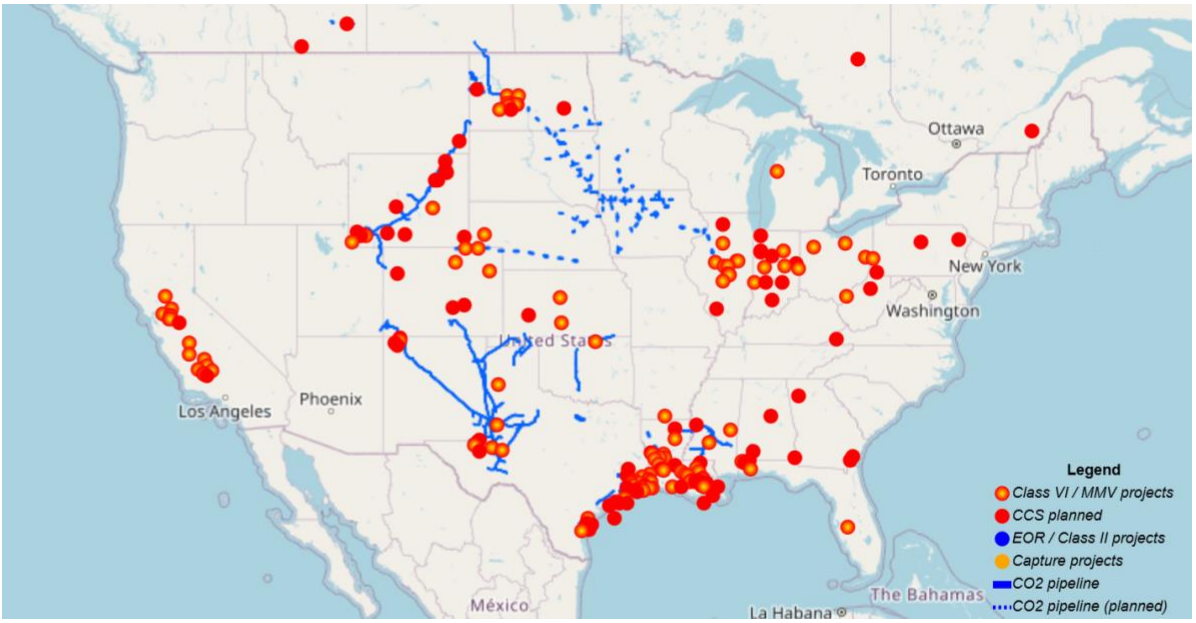

Figure 3: Map of potential CCS Class VI wells and planned CCS project in the US.[7]

Carbon Dioxide Removal (CDR) will be quantified using a rigorous Measurement, Reporting and Verification Protocol that will be developed after operation of the Pilot plant. However, certain principles will remain fundamental to protocol development. The amount of CO_2 sent to geological storage will be directly measured at the last, isolatable location. That is, the last location where the CO_2 can be attributed exclusively to Eleryc. If the CO_2 storage reservoir is storing CO_2 exclusively from Eleryc, then the last isolatable location will be at the well head immediately upstream of injection. If the storage location is common to numerous facilities or the CO_2 is routed to a common pipeline for transmission to the final reservoir, then the last isolatable location will be the point immediately prior to the Eleryc CO_2 mixing with other CO_2 streams. In the latter case, any fugitive emissions in the downstream pipelining of CO_2 will be attributed as storage losses (discussed below). The remaining elements of the CDR, including power demand and the emissions associated with power and materials, will be measured or calculated according to best practices.

Successful completion of the Demo plant will derisk the entire process at scale. The Demo plant will also enable hardening of the supply chain for all plant equipment and in-house fabrication which is the final technical hurdle to commercial roll out. At that point, Eleryc expects that plant scales will gradually be increased, through a combination of modularization, scaling, and parallel trains, to capacities of 1M TPA. 500 plants at this scale would allow achievement of Frontier's objective of 0.5 GTA of capture at costs below $100/tonne CO_2.

c. **Risks:** What are the biggest risks and how will you mitigate those? Include technical, project execution, measurement, reporting and verification (MRV), ecosystem, financial, and any other risks. 500-1000 words

The biggest risks to successful deployment of Eleryc's DAC process are:

1. Ever-changing Geopolitical Environment, both in terms of willingness to fund Direct Air Capture and in sourcing key components

[7] See https://www.ccusmap.com/markers/map/

2. General Risk of Scaling Hard-Tech
3. Widespread Availability of Low or Zero Carbon Energy
4. Durability of Geologic Sequestration
5. Air Contactor Development

These risks, which are common to all DAC processes to some extent, are those that can have the most significant impact on Eleryc's project timelines.

With regard to the geopolitical environment, we believe that the willingness to fund DAC projects is significantly correlated to the costs and risks of currently available DAC processes. To that end, providing a clear and demonstratable path to DAC processes that cost less than $100/tonne of CO_2 and use benign methods of capture will increase public willingness to pay for DAC deployment. The sourcing of key components will also be a risk, but we have already been in discussions with a supply-chain specialist who has developed sourcing models for similar systems. We believe this will not be a long-term impediment to deployment.

While we certainly acknowledge that the track record for scaling integrated hardware systems (i.e., hard-tech) is historically very challenging, we believe Eleryc is well positioned to succeed for several reasons. First, our chemistry is inherently very simple and proven – caustic solvents have long been known to be highly effective at capturing CO_2 as demonstrated by academia and industry alike, and the acid-base system we have selected avoids parasitic reactions that can sometimes plague other e-chem processes. Electrolyzer technology is inherently highly modular due to the repeated cell architecture and will benefit greatly from learning curves as we scale; likewise, both the chlor-alkali and hydrogen industries stand as proof points of the ability to achieve massive scale with electrolysis processes. Related to modularity and scaling, we're fortunate that the fundamental performance characteristics of e-chem cells scale gracefully, meaning that the performance demonstrated in our lab cells (~1/400th scale) is highly predictive of what's achievable in a production-scaled design, with good correlation observed thus far in lab testing vs our pre-pilot cells (~¼ scale). Additionally, our process substitutes a milder acid in place of commonly used HCl, and we operate at lower concentrations of caustic than chlor-alkali. This benefits us in two ways: (1) it helps us to avoid the need for costly or rare materials and exotic manufacturing processes in our cells, especially compared to other cleantech industries likes solar, batteries, and other (water) electrolyzers and (2) it avoids the need for dangerous chemical handling processes that drive overhead and complexity in our balance of plant. Notably, our process is a closed looping system with energy and water (just enough to make up for air contactor evaporative losses) as the only inputs, and CO_2 as the only output. While not producing revenue-able byproducts may be a drawback in the short term, we believe that in the long term this is actually a major benefit to scale as it means we are unencumbered by the need to find sufficiently scaled offtake partners that turn this into a liability. For example, while co-produced HCl could be a great revenue generator for a fledgling start-up at small scale, it would become a massive hazardous waste disposal problem at gigaton DAC scale. Finally, as a KV portfolio company, we have the benefit of standing on the shoulders of previous innovators in the e-chem space, learning from both their successes and failures alike. And, of course, we have assembled a team with wide experience developing, commercializing, and scaling hard-tech technologies, and while this doesn't guarantee success, we do believe it will help us avoid common pitfalls and maintain a realistic, achievable roadmap on our way to scale.

With regard to the cost and availability of low-carbon electricity, we believe that the continued deployment of large-scale renewable energy systems will grow in concert with our projected timelines. Moreover, Eleryc's DAC process can be responsive to changes in renewable power both in terms of power consumption and in terms of process turndown by adjusting the current density of our electrolyzer. We can run at higher current density when low-cost renewables are most available to maximize acid-base output and at lower current density when costs are higher to maximize efficiency. The Eleryc process also has options for operating the air capture system continuously, even without relying on large-scale battery storage options for renewables.

Additionally, it is worth discussing the permanence of CO_2 storage. We are committed to using the best storage solution available at a given site and our belief is that the best solution currently available for meaningful volumes of CO_2 will be Class VI injection wells. Permanence of such storage is obviously the critical feature of these sites, and we are committed to following best practices for monitoring losses from these wells. Additionally, should better storage options become available, we would pursue those as well.

Finally, we'd like to acknowledge our strategic choice thus far to focus our efforts on electrolyzer and chemistry development, rather than a whole plant approach. While we believe there are enough peer datapoints for successful air contactor implementation thus far to make this reasonably low risk for caustic based systems, we are still in the early days of our internal efforts to develop an air contactor, and much work remains to be completed. Thus, open technical and financial risk related to our air contactor does remain, which we've attempted to address currently with conservative scheduling in our roadmap and conservative capex estimates in our TEA modeling.

d. **Proposed offer to Frontier:** Please list proposed CDR volume, delivery timeline and price below. If you are selected for a Frontier prepurchase, this table will form the basis of contract discussions.

Proposed CDR over the project lifetime (tons) *(should be net volume after taking into account the uncertainty discount proposed in 5c)*	1,000 tonnes Pre-Purchase
Delivery window *(at what point should Frontier consider your contract complete? Should match 2f)*	2029-2030
Levelized cost ($/ton CO_2) *(This is the cost per ton for the project tonnage described above, and should match 6d)*	Please refer to non-public sections
Levelized price ($/ton CO_2)[8] *(This is the price per ton of your offer to us for the tonnage described above)*	$500/tonne CO_2

[redacted]

[8] This does not need to exactly match the cost calculated for "This Project" in the TEA spreadsheet (e.g., it's expected to include a margin and reflect reductions from co-product revenue if applicable).